1A. $(x, y) \in R1$ if and only if $xy = 4$.

R: Not reflexive because if $x = 4$, then $x*x = 16$.

A-R: Not anti-reflexive because if $x = 2$, then $x*x = 4$.

S: $ab = ba = 4$, therefore it is symmetric as (a, b) and $(b, a) \in R1$

A-S: Not anti-symmetric because if $a = 1$, $b = 4$, and $c = 1$, then ac equals 1.

T: Not transitive because there are real numbers that make statement (a, c) untrue.

1B. $(x, y) \in R2$ if and only if $x >= 2y$

R: Not reflexive because if $x = 4$, then $2(4)$ is bigger and only $x <= 2$ would work.

A-R: Not anti-reflexive because $x >= 2x$ is true when $x = 0$.

S: Not symmetric because x and y are not equal, as x is at least two times bigger than y.

A-S: Not anti-symmetric because when $x, y = 0$, symmetry is true.

T: Transitive because for both pairs of (a, b) and (b, c) to belong to R2, a has to be twice as big as b and b also has to be twice as big as c. Therefore, a will be at least 4 times bigger than c, meeting the requirement. Even if b or/and c equal 0, it will still validate.

2. $(x, y) \in R3$ if and only if $x/y \in Z$

R: For any x value (besides 0 which is impossible), x/x will equal 1. 1 belongs to Z, therefore it is reflexive.

A-S: $x/y = y/x \rightarrow x = y$. Therefore is anti-reflexive.

T: $(x/y)(y/z) = x/z \rightarrow (x, z)$. $x/z \in Z$, therefore is transitive.

R3 is a partial order.

3. $(x, y) \in R4$ if and only if $a = b + 3n$ for some integer n.

- Equivalence relation = reflexive, symmetric, and transitive.

R: When $n = 0$ and $(a, b) = (x, x)$, turns into $x = x + 3(0) \rightarrow x = x$. Therefore is reflexive.

S: Set a and b equal to each other. $b + 3n = a + 3n \rightarrow a = b$

T: $a = b + 3n$ and $b = c + 3n \rightarrow a = c + 6n$. 6n is still an int, therefore $(a,c) \in R4$ and is transitive.

EQUIVALENCE RELATION!

EQ of 2: Equivalence class of 2 translates into $a = 2 + 3n$

In set builder, $\{a \in R \mid a = 2 + 3n, n \in Z\}$

EQ of pi: Equivalence class of pi translates into $a = pi + 3n$

In set builder, $\{a \in R \mid a = pi + 3n, n \in Z\}$

4A. Let R5 be the relation on $Z+ \times Z+$ where $((a, b), (c, d)) \in R5$ if and only if $a/c = b/d$

R: (a, b) and (a, b). $a/a = b/b \rightarrow 1 = 1$, therefore it is reflexive.

S: $ad = bc \rightarrow c/a = d/b$. Therefore $(c, d), (a,b) \in R5$ and therefore is symmetric.

T: Show $(a, b), (e, f) \in R5$.

$(a, b), (c, d) \rightarrow a/c = b/d \mid (c, d), (e, f) \rightarrow c/e = d/f$

$(a/c)(c/e) = (b/d)(d/f) \rightarrow a/e = b/f \rightarrow (a, b), (e, f) \in R5$. Therefore is transitive

4B. The function f that fits the regulations of $f(a, b) = f(c, d)$ if and only if $((a, b), (c, d)) \in R5$ must be **f(x, y) = x/y**

$a/c = b/d \rightarrow ad = bc \rightarrow a/b = c/d$

Bigger relation can be split into smaller a/b functions.

4C.

Equivalence class containing $(1, 1) \rightarrow a/1 = b/1 \rightarrow a = b$

Therefore, if $a = b$, then the equivalence class with (1, 1) contains all integer pairs that have equal a and b values and are positive integers.

So, $\forall x$, $\forall y$ when $x = y$ and are pos. ints $\in$ R5 and are in equivalence class of (1, 1)

4D.

$[a, b] = \{(c, d) \mid a/c = b/d, c, d \in Z+\}$

Infinite amount of equivalence classes.

5A. $R6 = \{(f, g) \mid f(0) = g(0)\}$

R: $f(0) = f(0)$ and $f(1) = f(1)$, so $(f, f) \in R6$. Therefore, it is reflexive.

S: $(f, g) \in R6$, then $f(0) = g(0)$ and $f(1) = g(1)$. $g(0) = f(0)$ and $g(1) = f(1)$.

So, $(g, f) \in R6$ and therefore R6 is symmetric.

T: (f,g) and (g, h) both belong to R6. $f(0) = g(0)$ and $f(1) = g(1)$. Same goes with g, h.

$f(0) = g(0) = h(0)$ and $f(1) = g(1) = h(1)$. Since f(0 and 1) = h(0 and 1), $(f, h) \in R6$.

Therefore, it is transitive.

EQUIVALENCE RELATION!

Some infinite number of equivalence classes, as there is one where functions are equal, where functions have the same constant/zero value, and when functions have the same values for inputs.

5B. $R7 = \{(f, g) \mid \exists C \in Z, \forall x \in Z, f(x) - g(x) = C\}$

R: $(f, f) \in R7$ because $f(x) - f(x) = 0$ and $0 \in Z$. Therefore, it is reflexive.

S: $f(x) - g(x) = c \rightarrow -f(x) + g(x) = -c \rightarrow g(x) - f(x) = -c \in Z$

$(g, f) \in R7$, therefore it is symmetric.

T: (f, g) and (g, h) belong to R7.

$f(x) - g(x) = c$ and $g(x) - h(x) = j \rightarrow f(x) - g(x) + g(x) - h(x) = c + j$

$f(x) - h(x) = c + d \in Z \rightarrow (f, h) \in R7$. Therefore, it is transitive.

EQUIVALENCE RELATION!

Infinite amount of equivalence classes as for example, all functions with same non constant power will be in a class, all constant functions in a class, etc.

🎵 stay classy!